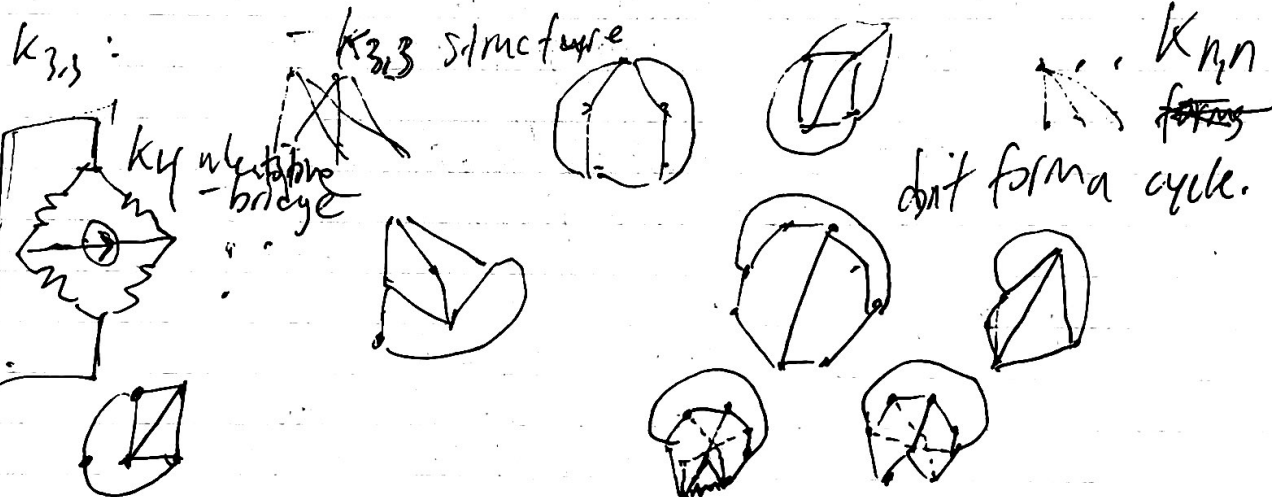
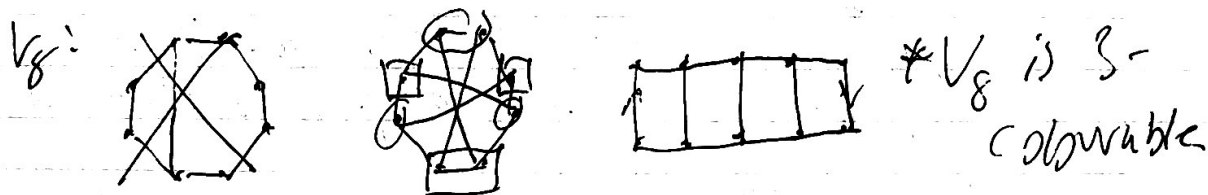


Feb 25 2026

Hadwiger's Conjecture - Amera

Theorem 2.1: For $0 \leq t \leq 3$, the graphs with no K_{t+1} minor are precisely the graphs that can be built by repeated clique sums, starting from graphs w/ at most t vertices.

Theorem 2.2: The graphs with no K_5 minor are precisely the graphs that can be built by repeated 0-, 1-, 2-, & 3- sums, starting from planar graphs and V_8 .



$\forall G \nRightarrow$ every graph w/ no K_5 minor is 4-colourable [HCC(4)].



By induction on the vertices of the graph by fact that a graph w/ no K_5 minor consists of at most a 3-sum of two graphs and V_8 is 3-colourable, and planar graphs are 4-colourable, by $\forall CT$.

2(2)

2(1) $t=0$, no K_1 minor $\Rightarrow$ an empty graph ; $0+0=0$
 $t=1$, no K_2 minor $\Rightarrow$ a graph w/ no cycles ;
 $t=2$, no K_3 minor $\Rightarrow$ a graph w/ no cycles parallel?

No $K_6 \Rightarrow 5$ colourable

• Look at structure

• Just start thinking/trying.

HC(5) [Seymour, Robertson, Thomas]

- w/o computer, w/o assume 4CT.

A smallest counterexample to $HC(5)$ has following structure:

- an apex graph ~~graph~~: a graph where a vertex deletion makes the graph planar.



let G be an apex graph.

- Any subgraph is 5-colourable?

- a minimal counterexample has no K_6 minor but is not 5 colourable, but any subgraph is 5-colourable.

Show that G is 6-vertex-connected; all vertices have $\deg \geq 6$ is shown in paper.

$\dots \Rightarrow G$ is apex

apex graphs are

$\rightarrow$ if not 6-vertex connected

- if it's the vertex: 5CT applies
- if not the vertex: not 5 colourable?

- No $K_5 \Rightarrow 10$ colourable

K -Degenerate graphs: a graph G if every non-empty subgraph (contains) has a vertex of degree at most k .

- forests are 1-degenerate
- series-parallel are 2-degenerate
- planar graphs are 3-degenerate

arity of planar graph

- a forest is finite graph graph w/ leaves. The leaves make it 1-degenerate
- a planar graph has an average of 4/3 degree, so

Any graph in which every vertex has a degree ≥ 3 contains a K_4 minor?

• series parallel graphs w/o K_4 minor

Any graph in which every vertex has degree ≥ 3 contains K_4 as a minor.

By induction:

Base case: K_4 

inductive step: I.H.: assume true for graph on n vertices.

let G be a graph on $n+1$ vertices, pick any vertex; any ~~two~~ edge incident w/ it and contracted gives a graph on n vertices w/ property that every vertex has degree ≥ 3 . By I.H., it contains a K_4 minor, so G has a K_4 minor.

• series-parallel graphs don't have K_4 minor [Dirac].

→ By contradiction, assume every graph has ≥ 3 edges (degree ≥ 3) ~~is a K_4 minor~~ so no 2-degenerate.

• by contradiction it has a K_4 minor, so series parallel graphs have no K_4 minor

— who Dirac proved series-parallel graphs are 2-degenerate

Hadwiger's Conjecture (II)
 $H(t) \Rightarrow$ every graph w/ no K_{t+1} minor is t -colourable
 - proved for $t \in [0, 5]$

$t=6$: Are graphs w/ no K_7 minor 6-colourable? OPEN Case

K_7^v is graph obtained by deleting 2 adjacent edges

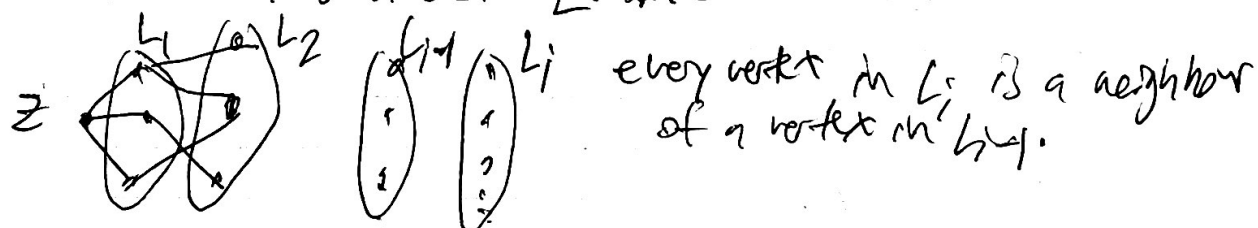
[Theorem 2025; Totschnig, Norin]: ~~K_7^v~~ Graphs w/ no K_7^v are 6-colourable.

Theorem [Alburt & Gonçalves 2013]: Every graph w/ no K_7 minor is 8-colourable, and every graph w/ no K_8 minor is 10-colourable.

[Wagner 1964]: All graphs w/ no K_{t+1} minor are 2^t -colourable

Proof: By induction on ~~size of the graph~~ t . Assume G is connected.

Fix some vertex z in G , and for each i let L_i be the set of vertices at distance i from z .



The subgraph induced at L_i has no K_t minor.

Contract all the vertices in $L_1, \dots, L_{i-1}$ w/ z into 1 vertex. The resulting vertex x is adjacent to all vertices of L_i .

If L_i contained a K_t minor, x w/ that minor would give a K_{t+1} minor; contradiction.

Since G has no K_{t+1} minor, then L_i has no K_t minor, so it is 2^{t-1} colourable, by induction.

Use one set of 2^{t-1} colours for the even levels (including $L_0 = \{2^0\}$) and one set for odd levels $\Rightarrow 2 + 2^{t-1} = 2^t$ colours.

• 2^{t-1} is an upper bound.

Degeneracy:

• G is k -degenerate, if every non-null subgraph has a vertex of degree at most k .

• forests: 1-degenerate

• series-parallel: 2-degenerate | no K_4 minor ~~no K_4 minor~~

• planar graphs: 5-degenerate

Theorem: If G is k -degenerate then its chromatic number is at most $k+1$.

Proof: If G has a vertex x of degree $\leq k$, delete x , $G-x$ is k -degenerate.

Base case: $k=0$ a null graph works w. 1 colour.

Induction:



Degeneracy gives another proof of $H(C(t))$ for $t \leq 3$ because for $t \leq 3$ every graph w/ no K_{t+1} minor is $(t-1)$ -degenerate (and hence $(t-1)+1 = t$ colourable).

7 *8

Theorem [Mader] 1967: for every graph H there exists C such that $|E(G)| \leq C|V(G)|$ for every graph G w/ no H minor.

But when $H = K_t$ for small values of t , we know the answer exactly

- for $n \geq 1$, n -vertex graphs w/ no K_2 minor have ^{at most} $n-1$ edges
- for $n \geq 2$, graphs w/ no K_3 minor have at most $2n-3$ edges
- for $n \geq 3$, graphs w/ no K_4 minor have at most $3n-6$ edges

~~if no K_t minor~~

Hadwiger's Conjecture III

- Find example of Kruskal's method for $H(C_5)$.
- G is k -degenerate if every non-null subgraph of G has a vertex of degree at most k .
- If G is k -degenerate, then $\chi(G) \leq k+1$.

Examples:

- Forests are 1-degenerate
- Series parallel graphs (no K_4 minor) are 2-degenerate.
- planar graphs are 3-degenerate
- For $t \leq 3$, graphs with no K_{t+1} -minor are t -degenerate; $\rightarrow t$ -colourable

Theorem [Mader 1967]: For every graph H there exists c such that $|E(G)| \leq c|V(G)|$ for every graph with no H minor ($\rightarrow \leftarrow$)

e.g.: $H = K_t$ for small values of t .

- i) for $n \geq 1$, n -vertex graphs with no K_3 minor (forests) have at most $n-1$ edges.
- ii) for $n \geq 2$, graphs with no K_4 minor has at most $2n-3$ edges
- iii) for $n \geq 3$, graphs with no K_5 minor has at most $3n-6$ edges
- iv) for $n \geq 4$, graphs with no K_6 minor has at most $4n-10$ edges.
- Every graph with no K_t minor has degeneracy at most $O(t(\log t)^{1/2})$

Ex) ... iv) $|E| \leq 4n-10$
 $\Rightarrow 2|E| \leq 8n-20$ $2|E| = \sum_{v \in V} \deg(v)$

so there exists a vertex v s.t. $\deg(v) \leq 7$

$\therefore 7$ degenerate is 8-colourable.

Suppose for a contradiction that every vertex has
degree ≥ 7 , then:

$$\sum_{v \in V} \deg(v) \geq \underbrace{7 + 7 + \dots + 7}_{|V|=n} \\ = 7n > 7n - 20$$

$\therefore$ contradiction.

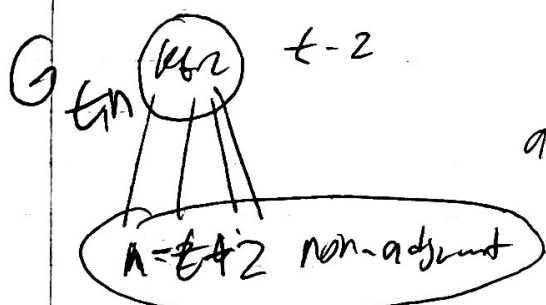
($\rightarrow \leftarrow$) $|E(G)| \leq c|V(G)|$, so there is a vertex of
degree $\leq \lfloor 2c \rfloor$, so $\lfloor 2c \rfloor$ -degenerate, so $(\lfloor 2c \rfloor + 1)$ -colourable.

March 19
2026

Hadwiger's Conjecture IV

Let $n \geq t-2$

Take the complete bipartite graph $K_{t-2, n-t+1}$ and add adjacency all pairs of vertices on the $t-2$ side.



every vertex in K_{t-2} is connected to all the vertices in $n-t+2$ vertices.

— n vertices in graph $G_{t,n}$

This Graph has $(t-2)n - \frac{(t-1)(t-2)}{2}$ edges

$$\begin{aligned} \# \text{ edges} &= (t-2)(n-t+2) + \frac{(t-1)(t-2)}{2} \\ &= (t-2)n - (t-2)^2 + \frac{(t-1)(t-2)}{2} \\ &= (t-2)n - (t-2) \left(\frac{2(t-2)}{2} - \frac{(t-1)}{2} \right) \\ &= (t-2)n - (t-2) \left(\frac{2t-4-t+1}{2} \right) \end{aligned}$$

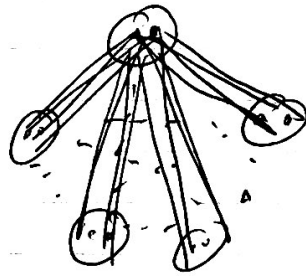
$$(*) = (t-2)n - \frac{(t-2)(t-1)}{2}$$

For $t \leq 5$, $G_{t,n}$ has maximum possible # of edges for a graph on n vertices w/o K_t minor

[Mader]: $t=6, 7$ has $(G_{t,n})$ number of edges as maximum # of edges.

- For $t=8$, pattern fails. $K_{2,2,2,2,2}$ has no K_8 minor but no satisfaction

$K_{2,2,2,2,2}$:



Theorem: For $t \leq 7$ and all $n \geq t-2$, every n vertex graph w/o K_t minor satisfies:

$$|E(G)| \leq (t-2)n - \frac{(t-1)(t-2)}{2}$$

$K_{2,2,2,2,2}$

$$\sum_{\text{vertices}} \deg(K_{2,2,2,2,2}) = \frac{80}{2} = 40 \text{ edges (Handshake Lemma)}$$

$$|E(G)| \leq (t-2)n - \frac{(t-1)(t-2)}{2}, \text{ let } t=8$$

$$= (6) \cdot 10 - \frac{7 \cdot 6}{2} = 39$$

- $K_{2,2,2,2,2}$ has no K_8 minor; greatest minor is K_7 . Take 1 vertex from each cluster

$t=3$: max # of edges $|E(G)| \leq (1)(1) - \frac{2 \cdot 1}{2}$

- graphs with n vertices w/o K_3, K_4, K_5 minors have $n-1, 2n-3, 3n-6$ edges

→ Substituting these graphs for $(t-2)n - \frac{(t-1)(t-2)}{2}$ gives some values.

to give counterexamples are known & well studied

Theorem [Jørgensen]: Let G be a graph on n w/
no K_5 minor, w/ $n \geq 6$ and

$\rightarrow |E(G)| \leq 6n - 21$ is isolated

let $|E(G)| = 6n - 20$ and G is built by pure
5-sums from copies of $K_{2,2,2,2,2}$

no edge
deletion can
change

Theorem [Song & Thomas]: Let G be a graph w/ no K_5
minor, w/ $n \geq 7$ and $|E(G)| \geq 7n - 28$, then $|E(G)|$
 $= 7n - 27$ and either $G = K_{2,2,2,3,3}$ or G can be built by
pure 6-sums from copies of $K_{1,2,2,2,2,2}$

As t grows (*) becomes completely isolated.

many
types
of
graphs

Identification

April 2 2026

Algebraⁿ for edges and vertices (Graphs) - prof. Wehrh

- elementary collapses are idempotent.

$$e^2 = e, \quad \text{if } e \subseteq s \text{ and } e \subseteq t$$

want idempotent with nothing above e , want equiv. class

- Nothing above e $s \geq e \Rightarrow e \leq s$ (maximal)

$\forall s \in S(n)$ (flow group) $\Rightarrow e \leq s$ and $e \leq s$
i.e. $e = s$
- maximal class.

$M = \{e\} \mid e \in S(n)$, By fact of $e = \tau_{xy}$
- an idempotent in a maximal class is a directed edge

let $e = f$ if $ef = e$ & $fe = f$, then $e = \tau_{xy}$
& $f = \tau_{yx}$ for $x \neq y$

$$\tau_{xy} \tau_{ab} = \tau_{xy} \text{ for } x \neq a$$

If $x \neq a$, $\tau_{xy} \tau_{ab} = \begin{cases} \tau_{xy} & \text{if } y \neq a, \neq \tau_{xy} \text{ (not an elementary collapse)} \\ g & \text{if } y \neq a, \neq \tau_{xy} \text{ (not an elementary collapse)} \end{cases}$

let $V =$

$$[e] =$$

- 12 are in equivalent class, if same source.
- no isolated vertices $\Rightarrow V$ is a source in bijection in V .
- vertices correspond to classes.

Find edges, and vertex set of a Graph

2 things have different sources $\Leftrightarrow$ classes are different
 an edge $\Rightarrow$ a pair of idempotents

What are the edges?

in Π if $\{x, y\}$ and edge in Π then $x \geq y$

$$\tau_{xy} \tau_{xx} \in M, \quad \begin{array}{l} \tau_{xy} \tau_{yx} = \tau_{yx} \\ \tau_{yx} \tau_{xy} = \tau_{xy} \end{array} \quad \left| \begin{array}{l} ef = f, e^2 = e \\ fe = e, f^2 = f \end{array} \right.$$

Conversely, If $e, f \in M$ and $ef = f$ and $fe = e, f \neq e$,
 where $e = \tau_{xy}$ and $f = \tau_{ab}$

$$ef = f \Rightarrow e = \tau_{ab} \text{ or } \tau_{b,c}$$

$$\boxed{\tau_{ab} \Rightarrow e = \tau_{b,c}} \quad \text{by corollary Fact 6.5.1 b}$$

edge iff

isomorphic to the flow semigroup,

- For every graph, a divisor looks like it came from K_{n-2} , but K_{n-2} is not a monoid

Tree-Width Theory - Amena

Tree Decompositions:

A tree decomposition maps a Graph G into a tree.

- * each node of Tree T is called a "bag" and it is a subset of $V(G)$.

A Tree Decomposition is denoted by $(T, \{B_i\})$
 - where $\{B_i\}$ is the collection of bags.

$\{B_i\}$ satisfies:

- 1) every vertex of G is in at least one bag B_i .
- 2) every edge (u, v) in G has both endpoints in at least one bag. $\exists i$ s.t. $u, v \in B_i$.
- 3) for every vertex v , the bags containing v form a connected subtree of T .

Tree widths:

The width of a decomposition $(T, \{B_i\})$ is $\max(|B_i| - 1)$.

* Tree width of a graph G is the minimum width among all possible tree decompositions of G .

Examples:

* G is a ~~cycle~~ tree:

$\Rightarrow$ Graphs w/ more bags have more information.

For every edge $e \in E(G)$ let B_e be the bag consisting only of the two end-vertices of e .

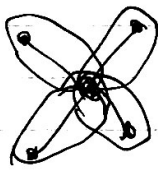
$\Rightarrow \forall e \in E, |B_e| = 2$, so $\max(|B_i| - 1) = 1$.

$\therefore$ The tree width is 1.

The Tree Decomposition: First, let $\tilde{G}$ be the graph obtained in the following way:

For any two vertices with bags B_e, B_f in the tree decomposition, we add an edge between B_e and B_f iff e and f have a vertex in common.

- There is a bag for every edge incident with v .



has a cycle $\rightarrow$ ~~not~~ $\Rightarrow$ not a tree.

Let T be a spanning tree of $\tilde{G}$

- Prof. Meunier says "Feels more like a covering." maybe not so algebraic.

Remark: ~~Thank you for~~ Have a good long weekend. I'm going to try ~~to apply~~ optimize PCBs w/ non-planarity for minimal amount of layers. Good luck w/ your research.

This was a very mind opening ^{guy} of months, hope your research breakthroughs

your research provided an ^{It is} really amazing experience ~~that is~~ good!

^{good, really} ~~It was~~ really ^{mind opening} inspiring, hope ~~all goes well~~ ^{and} you have a good ~~weekend~~ ^{weekend}

The tree decomposition $(T, \{B_v\})$

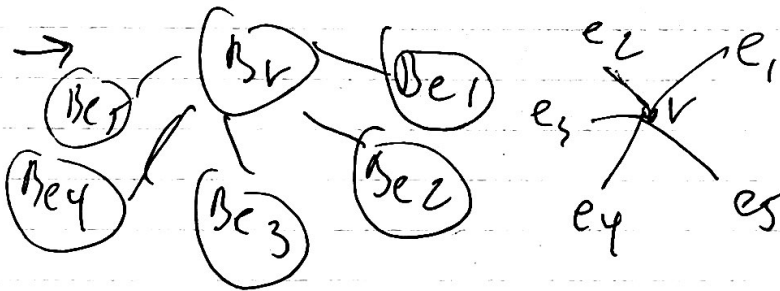
The tree decomposition $(T, \{B_i : i \in V \cup E\})$:

There are no edges between two bags of the same size (1 or 2).

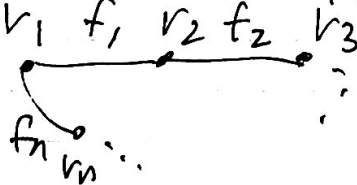
For any bag B_v for $v \in V$ and any bag B_e for $e \in E$, we have an edge in T between B_v and B_e iff v is an end-vertex of e .

Let $v \in V$, the bags containing v are B_v and all the B_e such that e is incident with v (v is end-vertex of e).

a star centred at B_v , so it is a subtree of T .



If there is a cycle $B_{v_1}, B_{f_1}, B_{v_2}, B_{f_2}, \dots, B_{f_n}, B_{v_n}$ in T , then



it gives a cycle in G , a contradiction.

T is connected. Let B_v and B_e be two bags in T . v is a vertex in G and $e = xy$ is an edge in G .

Since G is a ~~tree~~ tree, then there is a unique path in G between v and e . Say $v, f_1, x_1, f_2, x_2, \dots, x_n, e, y$.

Then $B_v, B_{f_1}, B_{x_1}, B_{f_2}, B_{x_2}, \dots, B_{f_n}, B_{x_n}, B_y$.